

Kombolcha Polytechnic College

Department of Information Technology

Group Project

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List of Acronyms

API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
CSS	Cascading Style Sheets

DFD	Data Flow Diagram
DLS	Digital Library System
DOC / DOCX	Microsoft Word Document formats
EPUB	Electronic Publication
ER	Entity-Relationship
FR	Functional Requirements
HTML	HyperText Markup Language
IT	Information Technology
JWT	JSON Web Token
MIME	Multipurpose Internet Mail Extensions
PDF	Portable Document Format
PPT / PPTX	Microsoft PowerPoint Presentation formats
PWA	Progressive Web App
RDBMS	Relational Database Management System
REST	Representational State Transfer
ROI	Return on Investment
SPA	Single-Page Application
SQL	Structured Query Language
SRS	Software Requirements Specification
UI	User Interface
UML	Unified Modeling Language

Abstract

Kombolcha Polytechnic College's traditional physical library faces operational challenges, including restricted operating hours, physical space limits, and single-copy book shortages. To

resolve these issues, this project presents the design and implementation of a web-based Digital Library System (DLS) providing 24/7 remote access to academic resources.

Built on a three-tier architecture using React, Node.js, Express, and PostgreSQL, the platform supports five distinct user roles: Administrator, Faculty Head, Librarian, Teacher, and Student. The system features hierarchical account provisioning, a multi-tiered document approval workflow (Faculty Heads approve Teacher uploads; Librarians approve general uploads), universal keyword search, and customizable file permissions, including an integrated online read-only viewer.

By modernizing resource distribution, the Digital Library System eliminates physical copy limits, streamlines search efficiency, and reduces long-term operational costs for the college.

Chapter One : Introduction

A college library system serves as the central hub for academic success, providing students and faculty with essential learning resources, research materials, and study support to drive scholarly achievement. Currently, Kombolcha Polytechnic College manages its library operations physically, relying on hardcopy books stored on shelves. However, this traditional setup faces major constraints, including limited physical book copies, restricted operational hours, and physical space limitations. To overcome these barriers and meet modern educational demands, the college is transitioning to a Digital Library System (DLS). By digitizing academic materials, the new platform provides 24/7 remote access to thousands of e-books and journals, enables instant keyword searching, and ensures every student can seamlessly retrieve study resources anytime and anywhere for their academic excellence.

1.1 Background of the Organization

Kombolcha Polytechnic College is one of the most popular polytechnic colleges found in Ethiopia, established in 1994 by the Ethiopian government. Located in Kombolcha city the home of many industries in South Wollo the college provides training across multiple technical professions, including Automotive, Garment, IT, Hotel and Tourism, and other vocational fields. To support its diverse hands-on training programs, academic research, and technical coursework, the college operates an institutional library for its students, instructors, and administrative staff.

Currently, the college relies on a traditional, fully manual library system where all physical books and technical manuals are stored on physical shelves. Under this model, students and instructors must be physically present inside the library during fixed operating hours to search for or borrow materials. Circulation records and catalog searches are maintained using paper ledgers. As student enrollment grows, this manual setup causes significant operational challenges, such as slow information retrieval, resource scarcity, book damage, and an inability to access learning materials outside of library hours.

To eliminate these constraints and support modern technical education, Kombolcha Polytechnic College is initiating the development of a Digital Library System. This new platform will digitize catalog management, enable electronic resource sharing (e-books, practical training modules,

and digital reference documents), and allow students and staff to search, reserve, and access academic materials remotely at any time.

1.2 Statement of the Problem

Kombolcha Polytechnic College currently relies on a traditional physical library system, where learning resources are stored as hardcopy books on physical shelves. This setup creates significant barriers to academic success for both students and faculty:

- Restricted Access & Hours: Students can only access reference materials during official library operating hours, preventing off-campus and late-night research.
- Resource Scarcity & Competition: Physical shelf capacity limits the total volume of books available. When multiple students require the same text, only a few can access it at a time, creating severe resource bottlenecks.
- Inefficient Resource Retrieval: Searching through physical stacks and manual index systems is time-consuming, often leading to missed or inaccessible study materials.
- Physical Space Constraints: The college's physical capacity cannot scale at the same pace as its growing student population and expanding academic curriculum.

Without a modern digital solution, students face unequal access to essential academic tools, directly impacting their learning outcomes, research efficiency, and overall academic achievements.

1.3 Objectives of the Project

1.3.1 General Objective

The primary objective of this project is to design and develop a web-based Digital Library System (DLS) for Kombolcha Polytechnic College that transitions the institution from a physical shelf-bound library to a digitized platform, ensuring seamless, 24/7 access to academic learning resources to enhance student educational success.

1.3.2 Specific Objectives

To achieve the general objective, the project will focus on the following specific targets:

- Centralize Digital Resources: Create a scalable cloud/server repository to host e-books, lecture materials, research journals, and reference documents in one accessible location.
- Enable 24/7 Remote Accessibility: Provide students and academic staff with continuous, off-campus access to learning materials from any device (smartphones, tablets, PCs).
- Implement Fast Search & Retrieval: Develop an intuitive search mechanism allowing users to filter and locate resources instantly by keyword, title, author, course, or subject category.
- Eliminate Resource Bottlenecks: Enable multiple users to access and read the same digital book concurrently, removing the limitations of single physical copies.
- Categorize Resources by Department: Organize academic materials according to the specific technical streams, departments, and modules taught at Kombolcha Polytechnic College for easy navigation.
- Improve Administrative Efficiency: Streamline administrative tasks by allowing library staff to upload, organize, update, and manage digital titles electronically with minimal manual overhead.

1.4 Feasibility Study

The feasibility study evaluates the overall practicality, technical viability, operational fit, cost-effectiveness, and legal compliance of developing the proposed Digital Library System (DLS) for Kombolcha Polytechnic College.

1.4.1 Technical Feasibility

The proposed Digital Library System is technically feasible as it relies on modern, proven, open-source web technologies and standard hardware infrastructure. The front-end interface is built using HTML5, CSS3, and Bootstrap to deliver a responsive grid layout across mobile devices, tablets, and desktop computers, while React powers a dynamic single-page application (SPA) capable of instant keyword search and real-time interface updates. On the back-end, Node.js combined with the Express framework provides a fast, asynchronous JavaScript runtime environment to efficiently process simultaneous REST API requests and serve digital documents. Database management is handled by PostgreSQL, an open-source object-relational database management system that ensures data integrity and fast execution when querying book metadata,

user accounts, and department categories. Furthermore, the system requires only standard server hosting—whether on a campus desktop server running Linux or Windows, or a basic cloud server—and mandates zero client-side installation since students and faculty access the application using standard web browsers like Chrome, Firefox, or Edge.

1.4.2 Operational Feasibility

Operationally, the Digital Library System aligns seamlessly with the daily routine and academic goals of Kombolcha Polytechnic College. By replacing physical shelf-bound search methods with a clean, browser-based search interface, the platform requires minimal user training and guarantees high adoption rates among students and instructors. Library staff transition from manual paper recordkeeping and shelf organization to an intuitive administrative dashboard where digital PDF titles can be uploaded, categorized, and managed electronically. Most importantly, providing round-the-clock remote access removes the physical opening-hour restrictions and limited seating capacity of the traditional library, giving students continuous access to study resources whenever and wherever they need them.

1.4.3 Economic Feasibility

From a financial perspective, the project offers exceptional value with minimal capital expenditure. The entire software stack—comprising React, Node.js, Express, Bootstrap, and PostgreSQL—consists of free, open-source technologies, eliminating software licensing fees entirely. Digitizing the library catalog significantly reduces long-term operational expenses associated with purchasing multiple physical hardcopies, rebinding damaged paper books, and expanding physical shelf space. Because a single uploaded digital file can serve hundreds of students concurrently without additional cost per reader, the system delivers a high return on investment while relying primarily on internal student development labor and routine server hosting costs.

1.4.4 Legal and Ethical Feasibility

The Digital Library System complies with standard institutional policies, ethical guidelines, and legal requirements. All e-books, lecture slides, and reference publications uploaded to the repository will strictly consist of institutional course materials, public domain texts, and open-

access academic publications. In addition, the system protects user privacy and data security by implementing encrypted password storage using industry-standard hashing algorithms, role-based user authentication, and secure database connections, ensuring a safe and compliant digital learning environment for Kombolcha Polytechnic College.

1.5 Scope and limitations of the Project

1.5.1 Scope of the Project

The scope of the Digital Library System defines the functional boundaries, access controls, resource governance, and administrative workflows built into the platform for Kombolcha Polytechnic College. The system establishes a structured operational hierarchy categorized into five distinct user roles: Administrator, Faculty Head, Librarian, Teacher, and Student.

- **User Management & Account Provisioning**: System registration operates strictly through role-based administrative provisioning rather than open self-registration. The Administrator logs into the portal to create accounts and credentials for Librarians and Faculty Heads. Each Faculty Head manages departmental staff by creating teacher accounts within their specific department. The Librarian creates and provisions account credentials for Students, while also overseeing student account records.
- **Profile Management**: All five user roles Administrator, Faculty Head, Librarian, Teacher, and Student have dedicated profile access, allowing them to view and update their personal account details after logging into the system.
- **Resource Upload & Multi-Tier Approval Workflow**: Content contribution is open to Administrators, Faculty Heads, Librarians, and Teachers. Students cannot upload documents directly, but they can submit formal resource requests for materials they need. To maintain repository quality and accuracy, all uploaded documents undergo an approval process before appearing in the main library catalog:
    - Materials uploaded by Teachers must be reviewed and approved by their respective Faculty Head.
    - Materials uploaded by other authorized roles are reviewed and approved by the Librarian prior to public listing.

- **Access Control & Reading Permissions**: Document uploaders can apply role-based access restrictions to limit visibility to specific target audiences or departments. Additionally, uploaders can enforce file permissions, deciding whether a resource is downloadable or restricted to read-only online viewing directly within the website viewer.

- **Resource Requesting & System Administration**: Students interact with the platform by searching, viewing, reading online, downloading permitted materials, and submitting resource requests. Librarians oversee daily library management, handle student resource requests, and curate the digital catalog.

1.5.2 Limitations of the Project

To preserve administrative oversight and prevent unauthorized system access, the project is bounded by the following restriction:

- It focuses mainly on digital resources and does not manage physical library inventory.
- **Restricted Student Self-Registration**: Students cannot register themselves directly through a public registration page. Every student user account must be created and registered by the Librarian before the student can log into the system.

1.6 Significance and Beneficiaries of the Project

1.6.1 Significance of the Project

The implementation of the Digital Library System (DLS) represents a transformative shift for Kombolcha Polytechnic College, moving its academic infrastructure from traditional physical shelf dependency to a modern digital platform. The significance of this project lies in several key impact areas:

- **Democratizing Learning Resources**: By providing 24/7 web access to e-books, lecture materials, and academic journals, the system eliminates physical library operating hour constraints, allowing students to study whenever and wherever they choose.
- **Resolving Resource Bottlenecks**: Physical libraries suffer from single-copy limitations where only one student can borrow a book at a time. Digitizing resources allows multiple

users to read and access the same material concurrently, eliminating wait times and book shortages.

- **Streamlining Academic Quality Control**: The multi-tier approval workflow (Faculty Heads reviewing Teacher uploads, Librarians reviewing general uploads) ensures that all digital materials hosted on the platform meet institutional quality standards and curriculum requirements.

- **Improving Operational & Search Efficiency**: Automated keyword search, category filtering, and online document reading cut resource retrieval time down from minutes of shelf searching to seconds, maximizing study and research efficiency.

- **Cost & Space Optimization**: The transition reduces the long-term institutional costs of purchasing, rebinding, and physically storing physical paper books, offering a scalable solution that grows alongside student enrollment.

1.6.2 Beneficiaries of the Project

The primary beneficiaries of the Digital Library System encompass all key stakeholder groups within the Kombolcha Polytechnic College community:

Students:

- Gain instant 24/7 access to required course materials, lecture notes, and e-books from smartphones, tablets, or PCs.

- Benefit from read-only online viewers and downloadable resources to suit off-campus study needs.

- Can submit formal resource requests directly to the library for missing or required study materials.

Teachers / Instructors:

- Easily upload supplementary reading materials, module guides, and reference documents directly to their respective department channels.
- Enforce role-based access restrictions and view/download permissions on their shared materials.

Faculty Heads:

- Maintain direct academic oversight by reviewing and approving instructional resources uploaded by department teachers.
- Easily manage departmental teaching staff accounts and ensure curriculum alignment.
- Librarians:
- Shift from manual paper tracking and physical shelf organization to an efficient digital dashboard.
- Process student account creations, review general document uploads, monitor student resource requests, and manage catalog metadata effortlessly.

College Administration & Institution:

- Modernizes the college's overall academic IT infrastructure, raising institutional standards.
- Maximizes resource ROI through open-source technology deployment (React, Node.js, PostgreSQL) without recurring software licensing costs.

1.7 System Development Approach & Methodology

1.7.1 System Development Approach

The development of the Digital Library System (DLS) utilizes the Waterfall Software Development Methodology. Waterfall provides a linear, sequential phase-by-phase framework where each stage Requirements Analysis, System Design, Implementation, Testing, Deployment, and Maintenance must be completed and documented before transitioning to the next.

This methodology was selected because the system requirements, user roles (Administrator, Faculty Head, Librarian, Teacher, and Student), access permissions, and document approval workflows for Kombolcha Polytechnic College are well-defined and fixed. Additionally, the Waterfall model enforces structured documentation and clear milestone reviews at every stage such as formalizing the Software Requirements Specification (SRS) and drawing system architecture diagrams prior to coding—making it ideal for academic system development and evaluation.

1.7.2 Methodology

To build an application that directly addresses the current manual challenges at Kombolcha Polytechnic College, a systematic data gathering and system design methodology was followed.

1.7.2.1 Data Gathering Methodology

Requirement gathering relied on two primary qualitative techniques:

- **Interviews with Library Staff**: Semi-structured interviews were conducted with the head librarian and administrative staff at Kombolcha Polytechnic College. These sessions provided essential insights into daily library operations, physical shelf management challenges, resource demand patterns, and the specific administrative rules required for student user registration and multi-tiered document approval pipelines.

- **Observation of Existing Physical and Digital Resources**: Direct observational study was conducted within the college library environment to analyze how physical books are categorized on shelves, how students search for study materials, and how digital learning resources (PDFs, modules, lecture slides) are currently distributed across departments.

This observational data directly informed the structure of the PostgreSQL relational database schema and the role-based uploader permission controls.

1.8 Development Tools and Technologies

The development tools, software frameworks, and documentation utilities selected for building and detailing the Digital Library System are outlined below:

Front-End Technologies:

- HTML5 & CSS3: Used to construct semantic web pages and establish custom visual styling and layouts.
- Bootstrap: Provides responsive, mobile-first grid components to ensure the portal adapts across desktop PCs, tablets, and smartphones.
- React: Powers the client-side single-page application (SPA), enabling fast dynamic search filtering and interactive UI components without forcing full page reloads.

Back-End Technologies:

- Node.js: Serves as the asynchronous, event-driven JavaScript runtime environment for building scalable backend services.
- Express.js: Provides the lightweight back-end web framework for designing RESTful API endpoints, routing client requests, handling file uploads, and enforcing role-based authentication middleware.

Database Technologies:

- PostgreSQL: Serves as the relational database management system (RDBMS) to securely store structured metadata, user credentials, department categories, and document access permissions with high data integrity.

Documentation & Design Utilities:

- Microsoft Word: Utilized for writing, formatting, and compiling the formal project proposal, Software Requirements Specification (SRS), and system manuals.

- Draw.io: Used for modeling system design diagrams, including Use Case diagrams, Data Flow Diagrams (DFD), Entity-Relationship (ER) diagrams, and system architecture flowcharts.

Chapter Two: Requirement Analysis

2.1 Introduction

Requirement analysis is a critical phase in the software development lifecycle that bridges the gap between high-level project goals and concrete technical execution. The primary purpose of this chapter is to define, specify, and analyze the functional, non-functional, business, and operational requirements for the proposed Digital Library System (DLS) at Kombolcha Polytechnic College.

This chapter translates the real-world operational challenges of the college's physical shelf-bound library such as limited opening hours, manual paper recordkeeping, single-copy book shortages, and lack of departmental access controls into a precise software specification. By establishing clear system boundaries, detailing user roles (Administrator, Faculty Head, Librarian, Teacher, and Student), and outlining multi-tiered document approval workflows, this analysis serves as the blueprint for system modeling, database design, and back-end REST API development using the selected technology stack (React, Node.js, Express, and PostgreSQL).

2.2 Current System Description

Kombolcha Polytechnic College currently relies on a traditional, manual physical library system. The physical library stores books, modules, and reference materials on physical shelves, organized manually by department and subject. Students and instructors must physically visit the library facility during official operating hours to search for, consult, or read academic resources.

Library staff manage day-to-day operations using paper registers, index cards, and manual ledger logbooks to track shelf inventory, catalogue physical books, and handle student inquiries.

2.2.1 Major Functions of the Current System

Despite its physical constraints, the existing system attempts to perform several core library management functions:

- Physical Book Storage & Shelf Organization: Books, research papers, and technical modules are physically categorized and placed on designated library shelves arranged by academic streams.
- In-Person Catalogue Search: Students and faculty search for required materials by walking through library aisles or asking library staff to manually inspect physical index files and ledger records.
- In-Library Reading Room Access: The physical library provides a designated reading hall where students can sit and study available reference books that are not permitted to leave the premises.
- Manual Recordkeeping & Inventory Management: Librarians manually record new book acquisitions, update physical ledger notebooks, and conduct periodical manual stock audits to inspect the physical condition of shelf inventory.

2.2.2 Problems of the Existing System

The current physical library system suffers from significant operational bottlenecks and limitations that hinder the academic progress of students and faculty:

- Time and Location Restrictions: Access to learning resources is strictly confined to official college working hours and physical library premises. Students cannot access study materials during evenings, weekends, or off-campus hours.

- Resource Shortages & Single-Copy Bottlenecks: Most physical textbooks and departmental modules exist in limited quantities. When a single book is in use by one reader, no other student can access that material, creating severe academic delays during project work and exam periods.

- Inefficient Resource Retrieval: Searching through physical stacks and manual ledger records is time-consuming. Students frequently waste valuable study time searching for misplaced, lost, or incorrectly shelved books.

- Physical Space and Seating Constraints: The library reading hall has a fixed seating capacity. As student enrollment at Kombolcha Polytechnic College grows, the physical facility cannot accommodate all students seeking a quiet study environment.

- Physical Wear, Tear, and Loss: Paper books and printed lecture modules degrade over time due to frequent handling, page tearing, moisture, and accidental misplacement, leading to permanent loss of valuable institutional knowledge.

- Lack of Centralized Digital Material Distribution: Instructors have no unified platform to share digital lecture slides, supplementary PDFs, or course modules directly with their department's students, resulting in fragmented and inconsistent material sharing.

2.3 Proposed System Description

The proposed Digital Library System (DLS) is a web-based repository engineered to modernize the library services at Kombolcha Polytechnic College. By migrating from a physical, shelf-bound operation to a centralized web platform, the system addresses the core bottlenecks of physical resource shortages, restricted opening hours, and manual recordkeeping. The platform provides a structured, secure, and user-friendly environment built on a modern open-source technology stack (React, Node.js, Express, and PostgreSQL).

2.3.1 Overview of Proposed System

The primary objective of the proposed Digital Library System is to centralize the acquisition, evaluation, storage, and distribution of educational and administrative resources across Kombolcha Polytechnic College. Unlike traditional repository systems with flat access structures, the proposed DLS operates under a strict, role-based architecture designed to maintain academic integrity, administrative accountability, and information confidentiality.

The proposed system replaces open public registration with a closed, multi-tier provisioning pipeline. System users are categorized into seven distinct roles, each assigned explicit operational boundaries and access permissions:

- **System Admin**: Maintains core system stability, configures global application settings, and provisions top-level administrative accounts.
- **Librarian Admin**: Functions as the chief operational administrator, overseeing library workflows and provisioning accounts for Librarians and Department Heads.
- **Librarian**: Manages routine cataloging, processes resource requests, issues college-wide announcements and provisions accounts for end-user patrons (Staff and Students).
- **Department Head**: Directs departmental academic content, provisions Teacher accounts within their respective department, and conducts primary reviews of teacher-submitted materials.
- **Teacher**: Authors and uploads instructional courseware, lecture slides, and laboratory guides, with configurable viewing and downloading parameters.
- **Staff**: Operates as institutional patrons with access to both general academic materials and restricted institutional/administrative documentation.
- **Student**: Operates as academic patrons with access restricted exclusively to approved educational materials and learning resources.

2.3.2 Operational Workflows & Access Control Security

1. Account Provisioning Chain

To safeguard system security and verify user identities, self-registration is disabled. User accounts are created through a top-down administrative chain:

- ❖ Top-Level Provisioning: The System Admin creates the Librarian Admin account.
- ❖ Administrative Delegation: The Librarian Admin provisions user credentials for Librarians and Department Heads.
- ❖ Operational Provisioning: Librarians register Staff and Students, while Department Heads register Teachers assigned to their academic department.

2. Multi-Tiered Content Approval Pipeline

To guarantee content quality and prevent unverified documents from entering the repository, all uploaded files must undergo formal verification:

- Departmental Verification: Educational materials uploaded by Teachers are routed to their respective Department Head for preliminary quality and curriculum alignment checks.
- Central Archival Validation: Content approved by Department Heads, as well as direct uploads from Librarians, undergo final validation by the Librarian Admin prior to publication in the public catalog.

3. Role-Based Document Access Control Matrix

The system enforces clear boundaries regarding document classification and accessibility: Academic vs. Institutional Document Classification: Uploaded resources are classified as either Academic Resources (textbooks, lecture notes, lab manuals) or Institutional Resources (administrative manuals, policy documents, governance reports).

- ✓ Staff Access Rights: Staff members are granted full catalog access, enabling them to search, view, and download (if it is downloadable) both academic materials and internal institutional documents.

- ✓ Student Access Rights: Students are restricted to academic resources only. The system automatically filters out internal institutional documents from student search indexes and catalog views.
- ✓ In-Browser Document Reader & DRM Controls: Uploaders can specify file-level security toggles:
- ✓ Downloadable: Allows patrons to read online or download local copies.
- ✓ Read-Only Viewer: Restricts patrons to an embedded, secure in-browser reader, preventing local file downloads to protect copyrighted or sensitive materials.

2.3.2 Functional Requirements

Functional requirements define the specific behaviors, operations, and services the Digital Library System (DLS) must execute. The system's functional capabilities across all seven roles are structured as follows:

1. System Administrator

- → Account Provisioning & Automated Delivery: Provision and manage top-level administrative accounts (Librarian Admin). Upon creation, the system automatically sends an email to the user containing their username, initial password, and assigned role.
- → Self-Service Password Reset**:** Initiate a "Forget Password" request using their registered email address to securely reset lost or forgotten credentials.
- → Category Management: Define, update, and manage global resource Categories and taxonomy frameworks used to classify all uploaded documents and materials.
- → Direct Resource Upload: Directly upload system-level documentation, policies, or global library assets.
- → Browse Library & Reading: Search, filter, and read all library catalog items, TTLM materials, and system documentation in the in-browser viewer.
- → System Monitoring & Security: Access system performance logs, security audits, and global platform settings.
- → Database Maintenance: Perform regular system updates, backup routines, and database optimization.

2. Librarian Administrator

- → Resource Request Fulfillment: Receive, review, and process formal Resource Requests submitted by students and staff, updating request statuses (Pending, In-Progress, Fulfilled).
- → Department Management: Create, configure, and manage institutional Departments to organize staff and student affiliations across the college.
- → User Account Management & Automated Credential Delivery: Provision, update, and manage accounts across the platform—including Librarians, Department Heads, Staff, and Students (excluding Teachers). The system automatically emails credentials (username, temporary password, and assigned role) directly to each newly registered user.
- → Self-Service Password Reset**:** Initiate a "Forget Password" request using their registered email address to securely reset lost or forgotten credentials.
- → Direct Resource Upload: Upload, classify, and publish core academic materials, TTLM modules, and reference texts directly into the central catalog.
- → Final Content Validation: Perform central validation and final approval on queued academic resources and TTLM documents submitted by Department Heads before publishing them to the public library catalog.
- → Browse Library & Reading: Browse, filter (by Department, Level, Category), and read all repository resources online.
- → Qualification Level Management: Oversee TVET qualification levels (Level I – V) and catalog metadata standards.

3 Department Head

- → Teacher Provisioning & Automated Credential Delivery: Exclusively provision and manage system accounts for Teachers within their assigned department. The system automatically emails registration details (username, initial password, and role) to the newly created Teacher account.
- → Departmental Resource Upload: Directly upload, classify, and configure access settings for department-specific TTLM and course materials.

- → Departmental Content Approval: Review and evaluate uploaded academic modules and TTLM materials submitted by departmental Teachers for curriculum accuracy before forwarding them to the Librarian Admin.
- → Browse Library & Reading: Search, filter, and read all published TTLM modules, academic resources, and departmental files via the in-browser document viewer.
- → Self-Service Password Reset**:** Initiate a "Forget Password" request using their registered email address to securely reset lost or forgotten credentials.

4 Teacher

- → Self-Service Password Reset**:** Initiate a "Forget Password" request using their registered email address to securely reset lost or forgotten credentials.
- → Automated Onboarding: Receive account login credentials (username, password, assigned role) automatically via email upon registration by the Department Head.
- → TTLM & Resource Upload: Upload digital course materials, lab manuals, and TTLM (Teaching, Training, and Learning Materials) in supported formats (e.g., PDF).
- → Metadata & Qualification Tagging: Assign precise metadata during upload, including assigned Category, module title, target Department, and Qualification Level (Level I through Level V).
- → Access Control Selection: Set DRM and viewing permissions (e.g., read-only vs. downloadable) for uploaded materials.
- → Browse Library & Reading: Browse, search by Department/Level/Category, and read approved TTLM modules and reference texts directly in the browser.
- → Submission Tracking: Track the status of submitted TTLM files through the multi-tier review pipeline (Pending, Approved, or Rejected).

5 Librarian

- → Patron Registration & Automated Credential Delivery: Register and manage day-to-day accounts for Students and Staff under the oversight of the Librarian Admin. Upon registration, the system automatically dispatches welcome emails containing login credentials and role details.

- → Self-Service Password Reset**:** Initiate a "Forget Password" request using their registered email address to securely reset lost or forgotten credentials.
- → Direct Content Management & Upload: Upload, index, and organize general library resources and reference books into defined system categories.
- → Browse Library & Reading: Search, filter, and view all cataloged materials online in the embedded document reader.

6 Student

- → Automated Onboarding: Receive system credentials (username, initial password, assigned role) directly via registered email upon creation.
- → Browse Library & Specialized Search: Browse and search the central repository using targeted filters, specifically filtering TTLM materials by Department (e.g., *IT*, *Automotive*), Category, and Qualification Level (e.g., *Level I – V*).
- → In-Browser Reading: Read approved digital resources and TTLM modules online via an embedded document viewer with enforceability of uploader DRM settings (e.g., download restrictions).
- → Self-Service Password Reset**:** Initiate a "Forget Password" request using their registered email address to securely reset lost or forgotten credentials.
- → Personal Organization: Bookmark relevant TTLM resources and view personalized reading history logs.
- → Resource Requesting: Submit formal requests for unavailable textbooks or specialized learning guides.

7. Staff

- → Automated Onboarding: Receive system credentials (username, initial password, assigned role) directly via registered email upon creation.
- → Self-Service Password Reset**:** Initiate a "Forget Password" request using their registered email address to securely reset lost or forgotten credentials.

→ Browse Library & Specialized Search: Browse, filter, and search general academic materials, TTLM modules (by Department, Category, Level), as well as internal institutional documents and policy guides.

→ In-Browser Reading: View document content directly online, save items to personal bookmarks, and track reading history.

→ Resource Requesting: Submit formal requests for additional institutional or academic reference materials.

2.3.3 Non-Functional Requirements

The non-functional requirements specify the quality attributes, performance benchmarks, and operational constraints of the system using the selected technology stack (React, Node.js, Express, and PostgreSQL):

- Performance & Response Time: The system must process database search queries and render filtered book catalog results in under 2 seconds. The Node.js and Express back-end must efficiently manage concurrent client REST API requests without significant latency.

- Usability & Responsiveness: Built with Bootstrap and React, the user interface must be fully responsive and mobile-friendly, adapting seamlessly across desktop monitors, tablets, and smartphone screens. The interface must follow clean, intuitive navigation standards to minimize user training requirements.

- Security & Access Control: Passwords must be encrypted using industry-standard password hashing (such as bcrypt) before being stored in the PostgreSQL database. Role-based middleware in Express must enforce strict endpoint authorization, ensuring users can only access features permitted by their specific role (e.g., preventing students from accessing upload endpoints).

- Availability & Reliability: The platform must maintain high availability (99% uptime) during active campus hours when hosted on a standard Linux/Windows local campus server or cloud instance, ensuring uninterrupted student access to study resources.

- Data Integrity & Scalability: The PostgreSQL relational database must maintain data consistency across user tables, resource metadata, and approval logs. The architecture must scale gracefully to accommodate growing student enrollment and expanding document repositories over time.

Chapter Three: System Model

The system model provides a structured graphical and narrative breakdown of how users interact with the Digital Library System (DLS) at Kombolcha Polytechnic College. By modeling system behaviors, actor interactions, and data flows, this chapter translates the functional requirements into clear visual designs for implementation.

3.1 Scenario

To demonstrate how the Digital Library System (DLS) functions in daily operations at Kombolcha Polytechnic College, the following real-world narrative scenarios describe end-to-end interactions across the system's seven institutional user roles and integrated repository features:

Scenario 1: Hierarchical Account Provisioning & Automated Email Credential Delivery

- At the beginning of the academic year, account credentials are provisioned strictly through a closed, top-down administrative chain to ensure security and identity verification without public self-registration. During registration, the system automatically dispatches an onboarding email containing the generated username, initial password, and assigned role to each user:
- System Admin: The System Admin creates and provisions top-level administrative accounts, including the Librarian Admin. Upon account creation, the system automatically emails login details and role assignments directly to the Librarian Admin's inbox.
- Librarian Admin & Department Head: The Librarian Admin configures college Departments and provisions system accounts for Librarians, Department Heads, Staff, and Students. Each user receives an automated system email containing their login credentials and role assignment. Upon receiving access, each Department Head provisions individual accounts exclusively for Teachers assigned to their specific academic department (e.g., Information Technology, Automotive, Garment). The system automatically emails each new Teacher their login details.
- Librarian: Simultaneously, the Librarian assists in managing day-to-day registrations for incoming Students and college Staff members. The system automatically dispatches welcome emails containing credentials and role assignments to all registered patrons.

Scenario 2: System Category Setup & Multi-Tier Content Upload Pipeline

- Upload access is restricted to administrative and teaching staff (System Admin, Librarian Admin, Department Head, Teacher, and Librarian)—Students and Staff do not have upload privileges. Content moves through a multi-tier verification workflow:
- Category Setup: The System Admin defines global resource Categories (e.g., Module, Reference Manual, Assessment, Institutional Policy) within the system taxonomy.

- Teacher Upload & Tagging: A Departmental Teacher uploads a new practical programming lab module in PDF format. During upload, the Teacher selects the assigned Category, tags the file with its target Department and TVET Qualification Level (Level I – V), and configures access permissions (e.g., online read-only vs. downloadable). The file enters a Pending status and automatically routes to the Department Head's review queue.
- Departmental Review & Central Catalog Publication: The Department Head evaluates the content for curriculum accuracy and approves it. The module then moves to the Librarian Admin for final validation. The Librarian Admin verifies metadata compliance and officially publishes the resource to the main library catalog under the dedicated TTLM directory. (Note: Direct uploads from System Admins, Librarian Admins, Department Heads, or Librarians follow direct publication paths upon administrative validation).

Scenario 3: Patron Search, TTLM Browsing, In-Browser Reading, and Central Request Fulfillment

- All seven system roles have full access to browse and read approved library resources using the embedded in-browser viewer, while interaction remains governed by role permissions:
- TTLM & Specialized Search Filtering: A Student, Staff member, or any other user logs into the portal using their emailed credentials and navigates to the Browse Library section. To locate specific Teaching, Training, and Learning Materials (TTLM), the user applies targeted filters:
- Department Filter: Selects their academic stream (e.g., Information Technology, Automotive, Garment).
- Category Filter: Filters by resource classification.
- Qualification Level Filter: Selects the target TVET qualification level (Level I through Level V).
- The search engine narrows down the repository to display only approved, level-appropriate TTLM modules and reference guides. (Note: Staff and administrative roles can additionally access restricted internal institutional documents and policy guides).
- Interactive In-Browser Reading & DRM Controls: Finding the required TTLM module, the user opens it directly within the embedded, in-browser document viewer for online

study. If the uploader designated the file as read-only, download controls remain strictly restricted. The user saves the module to their personal Bookmarks for quick revision and reviews their automatically logged Reading History.

- Resource Requesting & Central Admin Fulfillment: Needing an additional specialized textbook or TTLM guide not currently hosted in the repository, a student or staff member submits a formal Resource Request specifying the title, author, and subject. The request routes directly to the Librarian Administrator's administrative dashboard. The Librarian Admin reviews the request, updates its status (Pending, In-Progress, Fulfilled), and oversees the acquisition or digital upload of the requested material.

3.2 Use Case Model

3.3.1 Use Case Diagram

The Use Case Analysis provides a structural representation of how external users (actors) interact with the Kombolcha Polytechnic College Digital Library System (DLS) to perform core operational and administrative tasks. By modeling system behaviors from the user's perspective, this section defines functional boundaries, access controls, multi-tier approval workflows, and data processing routines across all institutional roles.

The system is designed around a closed, top-down administrative model paired with role-based authorization controls. Public or unauthenticated interactions are restricted to fundamental informational pages, whereas core operations such as multi-criteria TTLM (Teaching, Training, and Learning Materials) catalog filtering, DRM-enforced document viewing, hierarchical user provisioning, content curation, and resource request processing—require authenticated session access.

The following subsections outline the primary system actors, their operational permissions, and detailed specifications for key system use cases.

Actor	**Description**
System Admin	Manages top-level system configurations, global Resource Categories, and provisions Librarian Admin accounts. Has upload and reading access.
Librarian Admin	Manages college Departments, TVET Qualification Levels (Level I–V), user accounts (Librarians, Department Heads, Staff, Students *excluding Teachers*), final content approvals, and Resource Request fulfillment.
Department	Exclusively manages Teacher accounts within their department,

Head	reviews/approves teacher uploaded TTLM modules, and uploads departmental resources.
Teacher	Uploads TTLM and course materials, tags metadata (Category, Department, Qualification Level), sets DRM access permissions, tracks upload status, and reads library content.
Librarian	Registers day-to-day Student and Staff accounts, uploads general library assets, and browses/reads repository materials.
Student	Non-uploader role. Browses and reads TTLM materials (filtering by Department and Level I–V), manages personal bookmarks/history, and submits resource requests.
Staff	Non-uploader role. Browses and reads TTLM materials, academic texts, and internal institutional policies, manages personal bookmarks/history, and submits resource requests.
Guest	Unauthenticated user limited to viewing public static pages (Home, About, Developers).

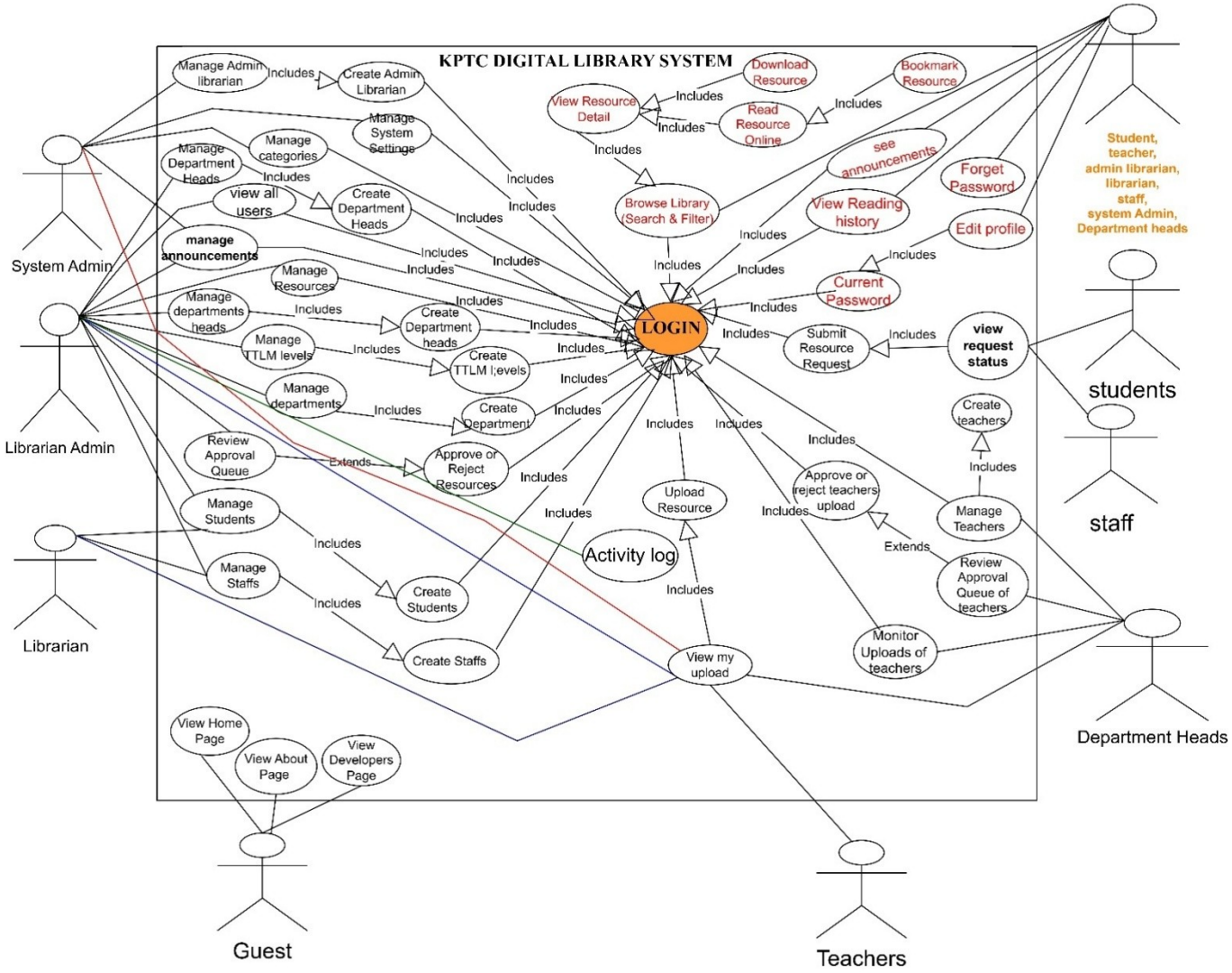


Figure 3.1 Usecase diagram

3.3.2 Use Case Diagram Description

High-Level Use Case Specifications

UC-01: Authenticate / Login & Automated Credential Delivery

- Primary Actors: All Users (System Admin, Librarian Admin, Department Head, Teacher, Librarian, Student, Staff)
- Pre-conditions: User account has been created by the designated administrator role.

Main Success Scenario:

- An administrator creates a user account.
- The system automatically dispatches an email to the user containing their username, initial password, and assigned role.
- User navigates to the login page and enters their credentials.
- System authenticates the user, verifies their role permissions, and grants access to their respective dashboard.

UC-02: Manage Categories & System Taxonomy

Primary Actor: System Admin

- Description: Define and maintain global classification categories for resource uploads (e.g., Module, Assessment, Policy).

Main Success Scenario: System Admin adds, updates, or deactivates resource categories used during file uploads.

UC-03: Manage Departments & TVET Qualification Levels

Primary Actor: Librarian Admin

- Description: Configure institutional departments (e.g., IT, Automotive, Garment) for user registration/affiliation and manage TVET Qualification Levels (Level I through Level V).

UC-04: Hierarchical User Account Management

Primary Actors: System Admin, Librarian Admin, Department Head, Librarian

Role Delegation Rules:

- System Admin: Provisions Librarian Admin accounts.
- Librarian Admin: Provisions Librarians, Department Heads, Staff, and Students (*strictly excludes Teachers*).
- Department Head: Exclusively provisions and manages Teachers assigned to their department.
- Librarian: Registers Students and Staff under Librarian Admin oversight.
- Post-condition: The system automatically sends login details to the created user's registered email.

UC-05: Multi-Tier Content Upload & Classification

- Primary Actors: System Admin, Librarian Admin, Department Head, Teacher, Librarian *(Students and Staff cannot upload)*

Main Success Scenario:

→ Author selects a file (PDF format) and inputs metadata: Title, assigned Category, target Department, and Qualification Level (Level I – V).

→ Author configures access control settings (e.g., Read-Only vs. Downloadable).

→ File is submitted. Teacher uploads move to Pending for Department Head review; Department Head approved files move to Librarian Admin for final catalog publication.

UC-06: Browse Library & TTLM Specialized Search

Primary Actors: All Authenticated Users (System Admin, Librarian Admin, Department Head, Teacher, Librarian, Student, Staff)

Main Success Scenario:

- User accesses the Browse Library interface.
- User applies multi-criteria search filters:
  - ✓ Department Filter (e.g., *Information Technology*)
  - ✓ Category Filter
  - ✓ TVET Qualification Level Filter (*Level I* through *Level V*)

- ✓ System displays matching approved TTLM modules and reference items.

UC-07: In-Browser Document Reading & DRM Access Control

Primary Actors: All Authenticated Users

Main Success Scenario:

- ➢ User selects a resource from the library catalog.
- ➢ The document opens in the embedded in-browser reader.
- ➢ The system enforces DRM policies set during upload (disabling download buttons if flagged read-only).
- ➢ The system automatically records the view in the user's Reading History. Users can also save the item to Bookmarks.

UC-08: Resource Request Submission & Central Fulfillment

Primary Actors: Student, Staff (Submitters) & Librarian Admin (Fulfiller)

Main Success Scenario:

- ❖ A Student or Staff member fills out a Submit Resource Request form specifying document details.
- ❖ The request routes directly to the Librarian Admin's central administrative dashboard.
- ❖ The Librarian Admin updates the request status (Pending, In-Progress, Fulfilled) and acquires/uploads the needed material.

Chapter 4: System Design

4.1 Introduction

The System Design phase translates the functional requirements and business rules specified in Chapter Three into a clear technical architecture. This chapter details the architectural models, database schema, data dictionary, key sequence diagrams, and user interface navigation flows for the Kombolcha Polytechnic College Digital Library System (DLS).

The structural design addresses the college's closed, top-down administrative workflow, incorporating:

- Role-Based Access Control (RBAC): Explicit boundary separation across seven primary roles (System Admin, Librarian Admin, Department Head, Teacher, Librarian, Student, and Staff).
- Automated Email Dispatch Engine: Seamless onboarding and identity delivery that automatically emails system credentials (username, temporary password, assigned role) upon account creation, alongside automated "Forget Password" reset routines for all registered users.
- Granular Taxonomy & TTLM Tagging: Database structures designed to index assets by global Resource Category (managed by System Admin), institutional Department, and TVET Qualification Level (Level I through Level V).
- Delegated Authority & Approval Pipelines: Database and workflow mechanics supporting multi-tier content approval (Teacher uploads queued for Department Head review, followed by final publication validation by the Librarian Admin).
- Centralized Request Fulfillment & DRM Viewing: Dedicated data models supporting central Resource Request handling by the Librarian Administrator, along with embedded, read-only DRM controls for non-uploader patron roles (Students and Staff).

4.2 Proposed Software Architecture

4.2.1 Software Architecture Overview

The Kombolcha Polytechnic College Digital Library System (DLS) is structured on a 3-Tier Modular Client-Server Architecture. This design pattern decouples the user interface, business logic, and database management layer, ensuring high system security, clear role separation, seamless scalability, and reliable processing of multi-tier approval workflows

1 Presentation Layer (Frontend Tier

The Presentation Layer handles all user interface rendering, user interactions, and client-side view management across standard web browsers:

- Role-Based Dynamic Dashboards: Renders custom navigation panels and administrative tools tailored to the authenticated user role (System Admin, Librarian Admin, Department Head, Teacher, Librarian, Student, or Staff).
- Embedded In-Browser Document Viewer: Displays uploaded PDF documents directly online for study while enforcing uploader-configured DRM permissions (e.g., hiding or disabling direct download controls on read-only TTLM files).
- Multi-Criteria Search & Filtering Interface: Provides specialized search tools for non-uploading patrons (Students and Staff) as well as academic staff to quickly locate TTLM content filtered by Department, Category, and TVET Qualification Level (Level I – V).

2 API Layer (Backend Tier - Entry Point)

The API Layer serves as the gateway for incoming network traffic between the client frontend and the backend services:

- RESTful Route Controller: Exposes structured HTTP endpoints (e.g., /api/auth/login, /api/resources/upload, /api/requests/fulfill) to process requests from the frontend.
- Request Parsing & Validation: Sanitizes incoming parameters, validates client inputs, handles session authorization headers, and routes validated requests to the underlying business logic layer.

3 Application Layer (Business Logic Tier)

The Application Layer processes all core rules, state transitions, security constraints, and automated workflows of the platform:

- ✓ Authentication & Role-Based Access Control (RBAC): Validates user credentials, manages authorization state, and enforces operational access boundaries across all system functions.
- ✓ Automated Email Dispatch Engine: Triggers automated background email tasks to deliver user credentials (username, temporary password, role assignment) upon account creation, while handling email-driven "Forget Password" recovery c for all user roles.
- ✓ Taxonomy & Metadata Module: Enforces metadata standards, governing global Categories (managed by System Admin), institutional Departments, and TVET Qualification Levels (Level I – V) (managed by Librarian Admin).
- ✓ Multi-Tier Content Approval Engine: Manages resource lifecycles by routing Teacher uploads into the Department Head review queue before advancing to the Librarian Admin for final catalog publication validation.
- ✓ Resource Request Management Engine: Directly routes resource request submissions from Students and Staff to the central administrative dashboard of the Librarian Admin for tracking and status updating (Pending, In-Progress, Fulfilled).

4 Data Access Layer

The Data Access Layer abstracts and manages communication between the business logic and the underlying storage mechanisms:

- ✓ Data Access Objects (DAOs) & Repositories: Encapsulates raw database queries (SQL/ORM operations), providing clean data manipulation methods (Create, Read, Update, Delete) for the application layer.
- ✓ Transaction & Storage Drivers: Manages database connection pools, handles data transactions, and coordinates file system reads/writes for uploaded PDF assets.

5 Database Layer

The Database Layer acts as the persistent physical storage foundation for all structured data and digital files:

- ❖ Relational Database System (RDBMS): Stores persistent records for user profiles, hashed security credentials, department affiliations, qualification levels, taxonomy categories, approval audit logs, user bookmarks, and reading history logs.
- ❖ Physical File Storage Repository: Securely stores uploaded PDF resources and teaching materials within structured server file directories, referencing their exact file paths inside the database tables.

4.2.2 Hardware/Software Mapping

Hardware/Environment	Mapped Software
Client Device	Web Browser, React 18 Frontend, React Router, Bootstrap 5, Axios, Browser Readers (PDF, EPUB, DOCX, Video)
Application Server	Node.js Runtime, Express.js, JWT Authentication, bcryptjs, Multer, Controllers, Models, PostgreSQL Connection Pool
Database Server	PostgreSQL 15+, Tables (users, resources, categories, bookmarks, reading_history, resource_requests, announcements, settings), Indexes, Foreign Keys, Constraints
File Storage Volume	PDF Files, EPUB Files, DOCX/PPTX Files, Video Files (MP4, MOV), Cover Images
Optional Conversion Runtime	LibreOffice/soffice, libreoffice-convert

Table 4.1: Hardware/Software Mapping

4.2.2 Access Control and Security

Key Elements of Access Control and Security

1. Authentication

Aspect	Description
Method	Email + Password Authentication
Process	User enters email/password → Backend verifies credentials using bcrypt.compare()

	→ JWT token generated on success → Token returned to frontend
Token Storage	JWT stored in browser localStorage
Token Usage	Included in Authorization header for subsequent API requests

Table 4.2: Authentication Security

2. Password Storage

Aspect	Description
Algorithm	bcryptjs
Process	Plain text password received → Salt generated → Password and salt combined and hashed → Hash stored in database
Security	Plain text password never stored; unique salt per password prevents rainbow table attacks

Table 4.3: Password Storage Security

3. Session Management

Aspect	Description
Method	JSON Web Tokens (JWT)
Token Content	Header (algorithm, type), Payload (user ID, role, expiration), Signature (generated with secret key)
Validation	Backend verifies signature and expiration on each request

Table 4.4: Session Management Security

4. Protected Routes

Aspect	Description
Middleware	requireAuth
Process	Check for Authorization header → If missing → 401 Unauthorized → If present → Verify JWT → If invalid/expired → 403 Forbidden → If valid → Attach user data to request → Proceed to controller

Table 4.5: Protected Routes Security

6. File Validation

Validation Type	Description
File Extension	Allowed: pdf, epub, ppt, pptx, mp4, mov
File Size	Maximum: Eg (100MB (configurable)
File Type	MIME type checked against declared type

Table 4.6: File Validation Rules

4.3 Database design

4.3.1 Entities and Attributes

Entity 1: USERS

Attributes	Data Type	Constraints	Description
full_name	VARCHAR(150)	NOT NULL	Full legal name of the user
email	VARCHAR(150)	UNIQUE, NOT NULL	Registered email address used for login and OTP password resets
password_hash	VARCHAR(255)	NOT NULL	Securely hashed user password
role	VARCHAR(20)	NOT NULL, CHECK (role IN ('admin', 'department_head', 'teacher', 'librarian', 'admin_librarian', 'staff', 'student'))	System authorization role
id_number	VARCHAR(50)	UNIQUE , NOT NULL	Student ID or Staff ID (required for student, department_head, and librarian)
department	VARCHAR(150)	NULLABLE	Department affiliation (used by department_head, teacher, and student)
expected_graduation	DATE	NULLABLE	Target graduation date for students (used for auto-deactivation)
created_by	INTEGER	FOREIGN KEY (users.id) ON DELETE SET NULL	Reference to the user who created this account

is_active	BOOLEAN	DEFAULT TRUE	Account active status flag
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the account was created
updated_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the account was last updated

Table 4.7: Entity Attributes: USERS

Entity 2: CATEGORIES

Attributes	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Unique auto-incrementing category identifier
name	VARCHAR(100)	UNIQUE, NOT NULL	Category name
description	TEXT	NULLABLE	Detailed description of the resource category
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the category was created

Table 4.8: Entity Attributes: CATEGORIES

Entity 3: RESOURCES

Attributes	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Unique resource identifier
title	VARCHAR(255)	NOT NULL	Resource document title
author	VARCHAR(150)	NULLABLE	Document author name
subject	VARCHAR(150)	NULLABLE	Subject area of the resource
keywords	TEXT	NULLABLE	Search keywords for indexing
description	TEXT	NULLABLE	Summary or abstract of the resource
category_id	INTEGER	FOREIGN KEY (categories.id) ON DELETE SET NULL	Reference to the associated category
file_path	VARCHAR(500)	NOT NULL	Storage path to the uploaded file on the server
file_size	BIGINT	DEFAULT 0	File size in bytes for display formatting

file_type	VARCHAR(20)	NOT NULL, DEFAULT 'pdf'	File extension format (e.g., pdf, epub, ppt, video)
cover_image	VARCHAR(500)	NULLABLE	File path to thumbnail cover image
visibility	VARCHAR(20)	NOT NULL, DEFAULT 'registered', CHECK (visibility IN ('public', 'registered'))	Visibility access scope
downloadable	BOOLEAN	DEFAULT FALSE	DRM control flag (FALSE = Read-Only online, TRUE = Downloadable)
allowed_roles	VARCHAR(20) []	NULLABLE	Restricted access array (NULL = all logged-in roles can access)
rejection_reason	TEXT	NULLABLE	Notes describing why a resource was rejected during approval
uploaded_by	INTEGER	FOREIGN KEY (users.id) ON DELETE SET NULL	Reference to the user who uploaded the file
status	VARCHAR(20)	NOT NULL, DEFAULT 'pending', CHECK (status IN ('pending', 'faculty_approved', 'approved', 'rejected'))	Multi-tier approval workflow status
approved_by	INTEGER	FOREIGN KEY (users.id) ON DELETE SET NULL	Reference to the Admin Librarian who provided final publication approval
is_ttlm	BOOLEAN	DEFAULT FALSE	Flag indicating if the resource includes a TTLM component
ttlm_level_id	INTEGER	FOREIGN KEY (ttlm_levels.id) ON DELETE SET NULL	Reference to the assigned TVET qualification level
view_count	INTEGER	DEFAULT 0	Total online view count
download_count	INTEGER	DEFAULT 0	Total download count
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the resource was uploaded
updated_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the resources record was last modified.

Table 4.9: Entity Attributes: RESOURCES

Entity 5: BOOKMARKS

Attributes	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Unique bookmark record identifier
user_id	INTEGER	FOREIGN KEY (users.id) ON DELETE CASCADE	Reference to the user creating the bookmark
resource_id	INTEGER	FOREIGN KEY (resources.id) ON DELETE CASCADE	Reference to the bookmarked resource
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the bookmark was created

Table 4.10: Entity Attributes: BOOKMARKS

Entity 6: READING_HISTORY

Attributes	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Unique history log identifier
user_id	INTEGER	FOREIGN KEY (users.id) ON DELETE CASCADE	Reference to the patron interacting with the resource
resource_id	INTEGER	FOREIGN KEY (resources.id) ON DELETE CASCADE	Reference to the accessed resource
action	VARCHAR(20)	NOT NULL, CHECK (action IN ('viewed', 'downloaded'))	Type of interaction performed by the user
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the interaction occurred

Table 4.11: Entity Attributes: READING_HISTORY

Entity 7: RESOURCE_REQUESTS

Attributes	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Unique resource request identifier
requested_by	INTEGER	FOREIGN KEY (users.id) ON DELETE CASCADE	Reference to the student or staff member submitting the request
title	VARCHAR(255)	NOT NULL	Title of the requested resource
author	VARCHAR(150)	NULLABLE	Author of the requested resource

subject	VARCHAR(150)	NULLABLE	Subject area of the requested resource
reason	TEXT	NULLABLE	Justification or explanation for requesting the resource
status	VARCHAR(20)	NOT NULL, DEFAULT 'open', CHECK (status IN ('open', 'fulfilled', 'declined'))	Status tracking resource request fulfillment
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the request was submitted

Table 4.12: Entity Attributes: RESOURCE_REQUESTS

Entity 8: ANNOUNCEMENTS

Attributes	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Unique announcement identifier
title	VARCHAR(255)	NOT NULL	Headline title of the announcement
message	TEXT	NOT NULL	Main message body of the announcement
posted_by	INTEGER	FOREIGN KEY (users.id) ON DELETE SET NULL	Reference to the user posting the announcement
is_active	BOOLEAN	DEFAULT TRUE	Publication status flag
image_filename	VARCHAR(255)	NULLABLE	File path for attached banner image
video_filename	VARCHAR(255)	NULLABLE	File path for attached video file
video_url	VARCHAR(500)	NULLABLE	External video URL link (e.g., YouTube)
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the announcement was created

Table 4.13: Entity Attributes: ANNOUNCEMENTS

Entity 9: SETTINGS

Attributes	Data Type	Constraints	Description
key	VARCHAR(100)	PRIMARY KEY	Unique key representing a system-wide configuration parameter
value	TEXT	NOT NULL	Value associated with the configuration key
updated_a	TIMESTAMP	DEFAULT	Timestamp when the setting was last

t		NOW()	updated

Table 4.14: Entity Attributes: SETTINGS

9. Entity: password_resets

Attributes	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Unique password reset request identifier
user_id	INTEGER	NOT NULL, FOREIGN KEY (users.id) ON DELETE CASCADE	Reference to the user requesting the password reset
otp_hash	VARCHAR(255)	NOT NULL	Securely hashed One-Time Password
expires_at	TIMESTAMP	NOT NULL	Expiration timestamp for the generated OTP
used	BOOLEAN	DEFAULT FALSE	Flag indicating whether the OTP has already been consumed
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the OTP was generated

Table 4.15: Entity Attributes: password resetting

10. Entity: ttlm_levels

Attributes	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Unique TVET level identifier
name	VARCHAR(100)	UNIQUE, NOT NULL	Level designation (e.g., Level 1, Level 2)
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the level was created

Table 4.16: Entity Attributes: TTLM levels

11. Entity: departments

Attributes	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Unique auto-incrementing department identifier

name	VARCHAR(150)	UNIQUE, NOT NULL	Institutional department name
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the department was created

Table 4.17: Entity Attributes: Departments

12. Entity: deleted_department_heads

Attributes	Data Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Unique log entry identifier
department_id	INTEGER	FOREIGN KEY (departments.id) ON DELETE CASCADE	Associated department identifier
department_name	VARCHAR(150)	NOT NULL	Name of the department led by the deleted head
full_name	VARCHAR(150)	NOT NULL	Full name of the deleted Department Head
email	VARCHAR(150)	NOT NULL	Email address of the deleted Department Head
id_number	VARCHAR(50)	NULLABLE	Identification number of the deleted Department Head
deleted_at	TIMESTAMP	DEFAULT NOW()	Timestamp when the account was deleted
deleted_by	INTEGER	FOREIGN KEY (users.id) ON DELETE SET NULL	Reference to the admin who deleted the Department Head account

Table 4.18: Entity Attributes: Deleted Department Heads

4.3.2 Entity Types (Strong vs Weak)

Entity Name	Classification	Key Dependency	Explanation
users	Strong Entity	Self-contained	Has its own primary key (id) and exists independently of all other entities
categories	Strong Entity	Self-contained	Exists independently with its own primary key (id)
departments	Strong Entity	Self-contained	Independent lookup table with its own primary key (id)
ttlm_levels	Strong Entity	Self-contained	Independent lookup table with its own primary key (id)
settings	Strong Entity	Self-	Independent configuration key-

		contained	value store with primary key (key)
announcements	Strong Entity	Non-identifying FK	Has an independent primary key (id) and continues to exist even if the poster is removed (ON DELETE SET NULL)
resources	Strong Entity	Non-identifying FKs	Has an independent primary key (id). Its foreign keys (uploaded_by, category_id, etc.) use ON DELETE SET NULL, meaning it exists independently of foreign entities
deleted_department_heads	Weak / Audit Entity	Identifying FK	Exists as an audit log dependent on departments.id (ON DELETE CASCADE)
password_resets	Weak Entity	Identifying FK	Security token bound directly to users.id (ON DELETE CASCADE)
bookmarks	Weak Entity (Junction)	Identifying FKs	Pure associative table bridging users and resources (ON DELETE CASCADE for both)
reading_history	Weak Entity (Log)	Identifying FKs	Activity log bound to both users and resources (ON DELETE CASCADE for both)
resource_requests	Weak Entity	Identifying FK	Patron request bound directly to users.id (ON DELETE CASCADE)

Table 4.19: Entity Types (Strong vs Weak)

4.3.3 Relationship Names and Cardinalities

Primary Entity (Parent)	Relationship Name	Related Entity (Child)	Cardinality (Parent : Child)	Foreign Key.
users	Creates / Onboards	users	1 : N (Optional)	created_by.
users	Submits	password_resets	1 : N	user_id.
users	Requests	category_requests	1 : N	requested_by.
users	Reviews	category_requests	1 : N (Optional)	reviewed_by.
users	Uploads	resources	1 : N	uploaded_by.
users	Approves (Stage 1)	resources	1 : N (Optional)	faculty_approved_by.
users	Approves (Final)	resources	1 : N (Optional)	approved_by.
users	Bookmarks	resources	M : N (via	user_id, resource_id.

			bookmarks)	
users	Views / Downloads	resources	M : N (via reading_history)	user_id, resource_id.
users	Submits	resource_requests	1 : N	requested_by.
users	Posts	announcements	1 : N	posted_by.
users	Deletes / Archives	deleted_department_heads	1 : N (Optional)	deleted_by.
departments	Logs Archive For	deleted_department_heads	1 : N	department_id.
categories	Classifies	resources	1 : N	category_id.
ttlm_levels	Tags Level For	resources	1 : N (Optional)	ttlm_level_id.

Table 4.20: Relationship Names and Cardinalities

4.3.4 Participation

Entity 1	Participation	Reason	Entity 2	Participation	Reason.
USERS (Creator)	Partial	User can exist without creating anyone	USERS (Created)	Partial	User can exist without being created (top-level Admin).
USERS	Partial	User can exist without requesting password resets	PASSWORD_RESETS	Total	Reset token cannot exist without an associated

					user.
USERS	Partial	User can exist without requesting categories	CATEGORY_REQUESTS	Total	Category request cannot exist without a requesting user.
USERS (Reviewer)	Partial	User can exist without reviewing category requests	CATEGORY_REQUESTS	Partial	Category request can exist before being reviewed.
USERS (Uploader)	Partial	User can exist without uploading resources	RESOURCES	Total	Resource must be uploaded by a user account.
USERS (Faculty Approver)	Partial	User can exist without giving Stage-1 approvals	RESOURCES	Partial	Resource can exist without requiring or receiving Stage-1 approval.
USERS (Final Approver)	Partial	User can exist without giving final publication approvals	RESOURCES	Partial	Resource can exist before receiving final approval.
USERS	Partial	User can exist without bookmarking resources	BOOKMARKS	Total	Bookmark cannot exist without referencing a valid user.
RESOURCES	Partial	Resource can exist without being bookmarked by any	BOOKMARKS	Total	Bookmark cannot exist without referencing a valid

		user			resource.
USERS	Partial	User can exist without viewing or downloading resources	READING_HISTORY	Total	History entry cannot exist without an associated user.
RESOURCES	Partial	Resource can exist without being viewed or downloaded	READING_HISTORY	Total	History entry cannot exist without an associated resource.
USERS	Partial	User can exist without submitting resource requests	RESOURCE_REQUESTS	Total	Resource request cannot exist without a requesting user.
USERS (Poster)	Partial	User can exist without posting announcements	ANNOUNCEMENTS	Partial	Announcement can remain active even if the posting user is removed.
USERS (Deleter)	Partial	Admin user can exist without deleting any department head	DELETED_DEPARTMENT_HEADS	Partial	Audit log entry can exist even if the deleting admin is removed.
DEPARTMENTS	Partial	Department can exist without having any deleted department	DELETED_DEPARTMENT_HEADS	Total	Audit log entry must always belong to a valid department.

		head logs			
CATEGORIES	Partial	Category can exist without having resources assigned to it	RESOURCES	Partial	Resource can exist without being assigned to a category.
TTLM_LEVELS	Partial	TTLM level can exist without having resources tagged under it	RESOURCES	Partial	Resource can exist without being tagged with a TTLM level.

Table 4.21: Entity Participation Types

4.3.4 ER Diagram

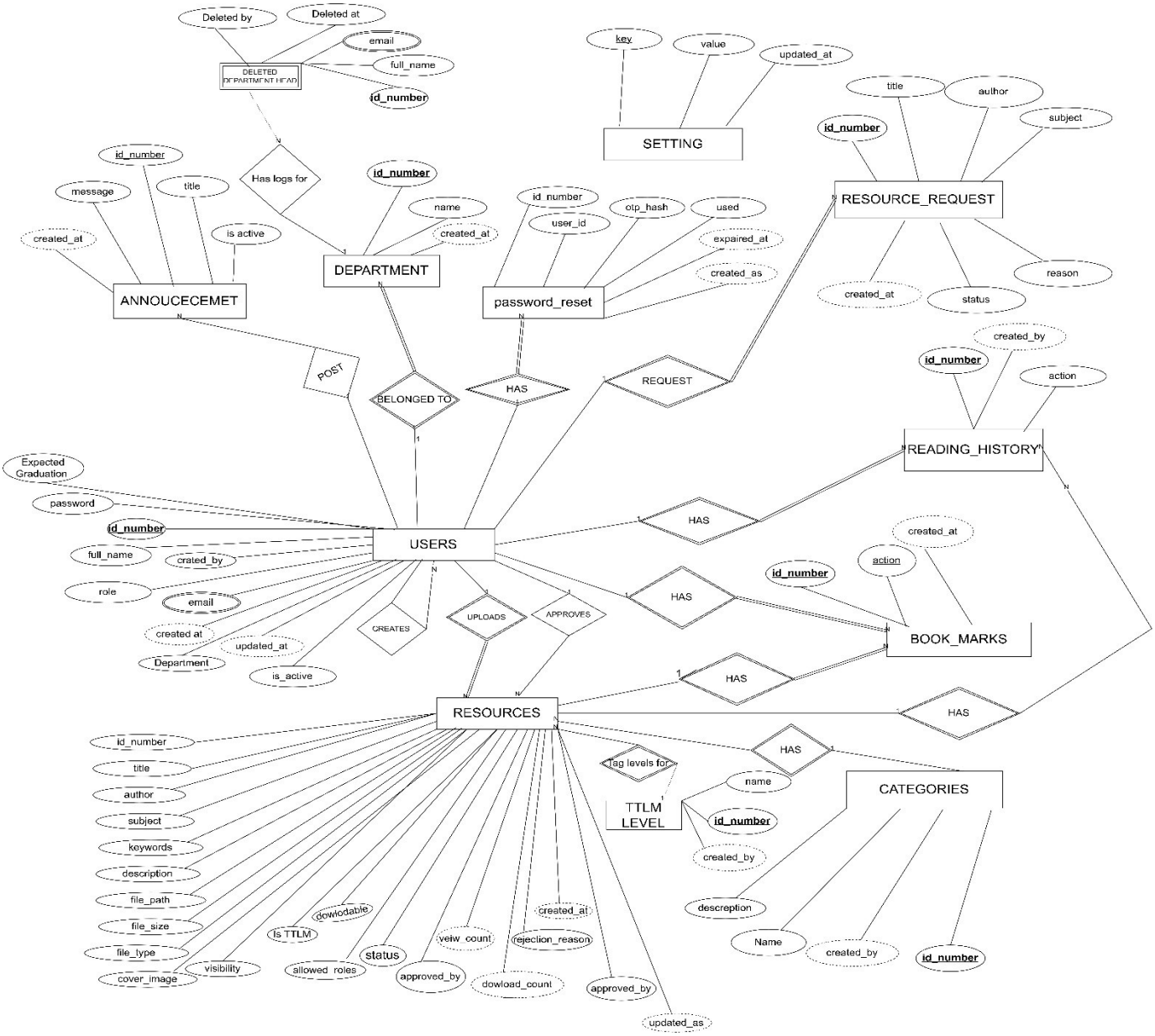


Figure 4.1: Entity-Relationship (ER) Diagram

Chapter 5: Conclusion and Recommendation

5.1 Conclusion

The development and implementation of the Digital Library System (DLS) mark a significant milestone in modernizing the academic infrastructure at Kombolcha Polytechnic College. By transitioning from a traditional, shelf-bound operation to a centralized web platform, the institution effectively resolves longstanding operational bottlenecks, including physical space limitations, restricted library operating hours, and single-copy resource shortages.

Built using a scalable three-tier architecture—leveraging a dynamic React frontend, a Node.js and Express backend, and a PostgreSQL database—the platform delivers continuous, round-the-clock remote access to e-books, lecture modules, and technical reference materials across all key user roles. The integration of structured access controls and a multi-tiered document approval workflow ensures that institutional resources remain secure while upholding rigorous academic quality standards. Ultimately, by replacing manual administrative processes with an open-source, digital solution, the project provides a cost-effective and sustainable framework that empowers both students and faculty in their academic pursuits.

5.2 Recommendation

To ensure the long-term sustainability, adoption, and growth of the Digital Library System at Kombolcha Polytechnic College, the following practical recommendations are proposed:

1. Registrar System Integration for Automated Self-Registration

Connect the Digital Library System directly with the college's official Registrar Database System. Allowing student account verification against registrar records will enable secure student self-registration while ensuring that only active, enrolled students gain access, significantly reducing the administrative workload on library staff.

2. Mobile Application and Offline Access Development

While the web platform is fully responsive and supports mobile viewing, future iterations should extend functionality through a Progressive Web App (PWA) or dedicated mobile application. This is recommended to deliver key features that web browsers cannot natively provide:

- Offline Study & Local File Caching: Enables students to save modules and e-books locally while connected to campus Wi-Fi, allowing continuous offline study at home without incurring mobile data costs or requiring steady connectivity.
- Real-Time System Push Notifications: Delivers instant OS-level push alerts to keep students updated when their resource requests are approved, new learning materials are uploaded, or urgent announcements are posted.

3. Targeted Stakeholder Training and Onboarding

Organize structured training workshops for librarians, department heads, and teaching staff to ensure smooth adoption. Training should focus on managing administrative dashboards, navigating the document approval pipeline, and setting appropriate file viewing and download permissions.

4. Infrastructure Expansion and Storage Optimization

As the digital catalog grows over time, the institution should invest in cloud backup solutions or dedicated high-capacity local servers to ensure data redundancy, high availability, and fast content delivery for large media files.